

Hassan A. Mahmoud

POSTDOCTORAL FELLOW

KAUST, Thuwal, Kingdom of Saudi Arabia

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Education

King Abdullah University of Science and Technology (KAUST)

Saudi Arabia

DOCTOR OF PHILOSOPHY IN MECHANICAL ENGINEERING

Fall 2025

- Dissertation: "Ultra-thin, Wireless and Flexible Sensors for Smart Structures"

Cairo University, Faculty of Engineering

Egypt

MASTER OF MECHANICAL DESIGN AND PRODUCTION

Fall 2021

- Thesis: "Analysis of Composite Bolted Connections under Out-of-Plane Loading"

Cairo University, Faculty of Engineering

Egypt

BACHELOR OF MECHANICAL DESIGN AND PRODUCTION

Fall 2017

- Thesis: "Reverse Engineering, Static and Dynamic Analysis of Turbocharger"

Research & Professional Experience

Postdoctoral Fellow

Saudi Arabia

MCEM LAB, KAUST

Nov. 2025 – Present

- Leading the development of fully printable wireless piezoelectric sensing platforms for structural health monitoring of mechanically demanding composite systems.
- Developed sensing methodologies for monitoring mechanical loading, environmental degradation, curing, and damage evolution in composite structures.
- Conducting in-situ testing of thin-ply CFRP composites using synchrotron X-ray computed tomography and digital volume correlation (DVC) to quantify internal deformation and damage evolution under mechanical loading.
- Developing sensing technologies for inspection of non-metallic pipelines under the *ENERCOMP* consortium in collaboration with *Saudi Aramco*.
- Contributing to the preparation and coordination of multidisciplinary research proposals targeting smart sensing and infrastructure monitoring.

PhD Student

Saudi Arabia

MCEM LAB, KAUST

Aug. 2021 – Nov. 2025

- Conducted interdisciplinary research integrating printed electronics, composite mechanics, and RF sensing to enable ultra-thin, passive, and wireless sensing platforms for structural health monitoring applications.
- **Ultrasensitive Cracked-based Wireless Passive Strain Sensor:** Developed ultrasensitive cracked-based passive wireless strain sensors using microfabrication technologies; developed a novel design pattern to control sensor sensitivity.
- **Fully Printable, Flexible, and Chipless Passive Wireless Strain Sensors:** Designed and fabricated flexible, fully printable wireless strain sensors with ultra-high strain sensitivity; optimized sensor linearity using variable spacing interdigitated electrode capacitive sensors; utilized e-tattoo transfer technology.
- **High Precision Wireless Strain Gauges for SHM** (collaboration with R&D Center, KAUST, Saudi Aramco): Developed highly sensitive, flexible, and on-demand wireless capacitive strain gauges; integrated with IoT system with wake-up radio for ultra-low power consumption; performed field tests at Aramco facilities.
- **Flexible Temperature Sensors using Laser-Induced Graphene (LIG):** Fabricated flexible thermoelectric sensors using processing-controlled Seebeck modulation in laser-induced graphene; implemented thermopile architectures to enhance thermoelectric sensitivity.
- **Flexible E-Tattoo Wireless Sensors for Smart Structures:** Designed integrable e-tattoo wireless strain sensors for real-time monitoring of composite structures; demonstrated in-situ monitoring of composite bonded joints during curing and service loading; developed low-power IoT-enabled sensing nodes for wireless data acquisition.
- **Enhanced Fracture Toughness of Composite and Hybrid Bonded Joints:** Performed research to enhance fracture toughness of adhesive joints using sacrificial cracks; developed a novel auxetic metal stiffener for metal-composite T-joints.
- **Wearable and Human-Machine Interface Sensing:** Developed stretchable capacitive sensors for gesture recognition in collaboration with *BrightSign Gloves* (UK); integrated sensing systems into wearable platforms for sign language monitoring.
- **Semantic-Native Event-Driven IoT Communication for SHM** (in collaboration with Qualcomm): Developed an AI-driven event-based architecture integrating Fast-SCNN edge inference with ultra-low-power wake-up radio sensors for on-demand sensing and transmission; validated for concrete crack detection with 90% lower data transmission, 96.8% communication reliability, and >3× longer sensor lifetime than conventional duty-cycled IoT systems.

R&D Intern

Saudi Arabia

EXPEC ARC, SAUDI ARAMCO

Jun. 2025 – Aug. 2025

- Conducted collaborative research with the advanced sensing team at KAUST Upstream Research Center (KURC), EXPEC Advanced Research Center (ARC) on sensing technologies for monitoring subsurface and energy infrastructure.
- Developed and experimentally validated a wireless capacitive sensing platform for low-frequency seismic signal detection.
- Demonstrated high sensitivity within seismic frequency ranges relevant to subsurface exploration applications.

Graduate Research Assistant

Egypt

CENTER FOR ADVANCED MATERIALS (CAM), THE BRITISH UNIVERSITY IN EGYPT (BUE)

Jun. 2019 – Jul. 2021

- Conducted multidisciplinary research in materials mechanics, composite structures, and renewable energy systems while contributing to student supervision and laboratory activities.
- **Wind Turbine Design:** Designed and analyzed a 5 kW vertical-axis and 10 kW horizontal-axis wind turbine using FEM (ABAQUS); supervised composite blade manufacturing via vacuum infusion; the manufacturing of the mechanical components, the assembly process, and field testing.
- **Optimization of Bandpass Absorption Glass Filters:** Investigated bandpass absorption glass filters through compositional optimization using rare-earth dopants; fabricated and characterized glass filters to evaluate photovoltaic efficiency impact.

Undergraduate Research Assistant

Egypt

MECHANICAL VIBRATIONS & ROTOR DYNAMICS LAB (PROF. ALY EL-SHAFEI'S LAB), CAIRO UNIVERSITY

Sep. 2016 – Jun. 2017

- Performed high-resolution 3D scanning of turbocharger components and reconstructed CAD geometries using Geomagic Design X for structural and fluid analyses.
- Conducted elemental and microstructural characterization using energy-dispersive X-ray spectroscopy (EDX).
- Performed CFD and fluid–structure interaction simulations of turbine and compressor rotors using ANSYS.

Technical Expertise

Modeling & Simulation	Finite element and multiphysics simulation in ABAQUS (structural, thermal, composites, fracture/damage mechanics) and ANSYS (mechanical, thermal, Maxwell); electromagnetic/RF modeling in HFSS and CST Studio; circuit-level and multiphysics modeling in LTSpice and COMSOL; solid modeling and geometry processing in SolidWorks, Geomagic Design X, Avizo, and Clewin
Scientific Computing	MATLAB and Python for data analysis, multiphysics modeling, and sensor signal processing; LaTeX, Inkscape, and KeyShot for scientific documentation and figure preparation
Fabrication & Prototyping	Composite manufacturing (hand layup, vacuum infusion, prepreg processing); micro/nanofabrication (sputtering, electroplating, wet etching, plasma RIE, photolithography, UV alignment, laser direct writing); rapid prototyping and printed electronics (polymer/SLS 3D printing, screen printing, dispenser printing with Voltera NOVA, waterjet/laser cutting)
Characterization	Mechanical testing (material characterization, vibration, bulge test, nanoindentation, Split Hopkinson Pressure Bar test); Fracture and damage characterization, In-situ testing and characterization (SEM, X-ray CT scan, X-ray synchrotron, DIC, DVC); imaging/materials characterization (CT, SEM, optical microscopy, Raman spectroscopy, DSC, FTIR, contact/non-contact profilometr); electrical/RF characterization (VNA, LCR meters, impedance analyzers, probe stations, nanovoltmeters, four-point probe)
Languages	Arabic (native), English (proficient)

Grants & Research Funding

High Precision Wireless Gauges for Structural Health Monitoring

Total: \$191,814

ENERCOMP CONSORTIUM (KAUST & SAUDI ARAMCO)

2024

RTF – Stretch Sensors for Integration into Smart Gloves

Total: \$751,600

KAUST RESEARCH TRANSLATION GRANT

2022

Honors & Awards

2026	EWSHM 2026 Students and Young Professional Challenge Awards , EWSHM 2026	France
2025	Honorary Certificate of Appreciation (Top 20 in 2025 ComSoc Student Competition) , IEEE ComSoc 2025	Taiwan
2025	Top PhD Student Ambassador at KAUST , Qualcomm	KSA
2025	Outstanding Performance Award, EXPEC ARC KURC , Saudi Aramco	KSA
2025	PSE Dean's Research Award , KAUST	KSA
2024	PSE Dean's Research Award , KAUST	KSA
2024	Best Poster Award, LOPEC 2024 , LOPEC	Germany
2022	Recognition Award of Engineering Achievements , The British University in Egypt	Cairo, Egypt

Publications

JOURNAL PUBLICATIONS – PUBLISHED

1. **Hassan A. Mahmoud**, O. Khalifa, Y. Shi, A. Wagih, T. Y. Al-Naffouri, G. Lubineau, "Integrable E-Tattoo Wireless Sensors for In-Situ Health Monitoring of Composite Structures," *Sensors and Actuators A: Physical*, 2026. DOI: 10.1016/j.sna.2026.118364
2. N. Divakaran, Y. Kara, **Hassan A. Mahmoud**, E. Hill, G. Lubineau, "Frequency-Tunable Piezocapacitive PDMS/MWCNT Nanocomposite Stretchable Sensors for Sign Language Recognition.," *Materials & Design*, 2026. DOI: 10.1016/j.matdes.2026.116194
3. A. Abdou, **Hassan A. Mahmoud**, M. Al Refae, Y. Zhang, H. Rekik, A. Seno, G. Lubineau, "Leveraging Process-Induced Electromechanical Anisotropy of Laser-Induced Graphene for One-shot Engraving of Highly Sensitive Strain Sensors," *ACS Applied Nano Materials*, 2026. DOI: 10.1021/acsanm.5c05242
4. W. Badeghaish, A. Wagih, **Hassan A. Mahmoud**, S. Abdel-Monsef, P. Maimi, G. Lubineau, "Effect of Supercritical CO₂ Aging on the Microstructure and Fracture Properties of 3D Printed PEEK," *Journal of Materials Research and Technology*, 2025. DOI: 10.1016/j.jmrt.2025.12.085
5. A. Wagih, F. E. Oz, R. Melentiev, **Hassan A. Mahmoud**, P. Maimi, M. Abdelaziz, G. Lubineau, "Hydrogen Storage and Transportation Polymeric Tanks: Damage, Permeation Mitigation, and Sensing," *International Journal of Hydrogen Energy*, 2025. DOI: 10.1016/j.ijhydene.2025.151334
6. **Hassan A. Mahmoud**, A. AlJafari, M. Bahabri, M. A. Alrefae, G. Lubineau, "Processing-Controlled Seebeck Modulation in Laser-Induced Graphene for Flexible Temperature Sensors," *ACS Applied Electronic Materials*, 2025. DOI: 10.1021/acsaelm.5c00562
7. **Hassan A. Mahmoud**, H. Nesser, G. Lubineau, "A Fully Printable Strain Sensor Enabling Highly-Sensitive Wireless Near-Field Interrogation," *Advanced Science*, 2025. DOI: 10.1002/advs.202411346
8. G. Lubineau, M. Alfano, R. Tao, A. Wagih, A. Yudhanto, Z. Li, K. Almuhammadi, M. Hashem, P. Hu, **Hassan A. Mahmoud**, F. Oz, "Harnessing Extrinsic Dissipation to Enhance Toughness of Composites and Joints," *Advanced Materials*, 2024. DOI: 10.1002/adma.202407132
9. **Hassan A. Mahmoud**, H. Nesser, A. Wagih, G. Lubineau, "Optimization Fragmentation of Metallic Film for Cracked-Based Strain Sensors," *ACS Applied Electronic Materials*, 2024. DOI: 10.1021/acsaelm.4c00819
10. A. Wagih, **Hassan A. Mahmoud**, G. Lubineau, "3D Printed Auxetic Stiffener for Metal-Composite T-Joints," *Materials & Design*, 2024. DOI: 10.1016/j.matdes.2024.112963
11. A. Alshedayfat, A. Wagih, A. Yudhanto, **Hassan A. Mahmoud**, G. Lubineau, "Mode II Fatigue Characteristics of a Composite Bonded Joint with Microstructured Adhesive Bondline through Tailored Sacrificial Cracks," *Composites Part A: Applied Science and Manufacturing*, 2024. DOI: 10.1016/j.compositesa.2024.108090
12. A. Wagih, H. Junaedi, **Hassan A. Mahmoud**, G. Lubineau, A. Kumar, T. A. Sebaey, "Enhanced Damage Tolerance and Fracture Toughness of Lightweight Carbon-Kevlar Fiber Hybrid Laminate," *Journal of Composite Materials*, 2024. DOI: 10.1177/00219983241235853
13. H. Nesser, **Hassan A. Mahmoud**, G. Lubineau, "High-Sensitivity RFID Sensor for Structural Health Monitoring," *Advanced Science*, 2023. DOI: 10.1002/advs.202301807
14. A. Wagih, **Hassan A. Mahmoud**, R. Tao, G. Lubineau, "Towards Tough Thermoplastic Adhesive Tape by Microstructuring the Tape Using Tailored Defects," *Polymers*, 2023. DOI: 10.3390/polym15020259

CONFERENCE PROCEEDINGS

1. **Hassan A. Mahmoud**, O. Khalifa, Y. Shi, A. Wagih, T. Y. Al-Naffouri, G. Lubineau "Printed E-Tattoo Wireless Sensor for Smart Monitoring of Composite Structures" *Proceedings of the 12th European Workshop on Structural Health Monitoring (EWSHM 2026)*
2. **Hassan A. Mahmoud**, O. Khalifa, A. H. Seno, M. A. Hussaini, N. Kouzahya, A. Asiri, A. Al-Jarro, A. shehri, H. Saiari, T. Y. Al-Naffouri, G. Lubineau "Capacitive Wireless Strain Sensors for Structural Health Monitoring of Composite Pipes" *Proceedings of the 12th European Workshop on Structural Health Monitoring (EWSHM 2026)*
3. **Hassan A. Mahmoud**, G. Lubineau, "Advanced E-Tattoo Wireless Strain Sensors for Smart Structural Health Monitoring of Composite Structures," *Proc. SPIE, Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2025*, vol. 13435, 134350D, 2025. DOI: 10.1117/12.3052549
4. **Hassan A. Mahmoud**, G. Lubineau, "Flexible Temperature Sensor using Laser Induced Graphene (LIG) Based on Processing-Controlled Seebeck Contrast," *Proc. SPIE, Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2025*, vol. 13435, 1343517, 2025. DOI: 10.1117/12.3050755

5. **Hassan A. Mahmoud**, H. Nesser, G. Lubineau, "Integrated Piezoresistivity and Frequency Modulation for High Sensitivity Printed Capacitive Strain Sensing," *IEEE FLEPS 2024*, DOI: 10.1109/FLEPS61194.2024.10604024
6. **Hassan A. Mahmoud**, H. Nesser, G. Lubineau, "Advancing Wireless Strain Sensing with Low-Cost Printable Supersensitive Radiofrequency Patches," *e-Journal of Nondestructive Testing*, 2024. DOI: 10.58286/29725
7. **Hassan A. Mahmoud**, H. Nesser, A. Wagih, G. Lubineau, "High Sensitivity Wireless Strain Sensor for SHM Applications," *Structural Health Monitoring* 2023, DOI: 10.12783/shm2023/36847
8. G. Lubineau, **Hassan A. Mahmoud**, H. Nesser, "Toward Smart Composites for Structural Health Monitoring via Highly Sensitive Capacitive Wireless Sensor," *ECCOMAS Thematic Conference on Smart Structures and Materials (SMART 2023)*, DOI: 10.7712/150123.9766.443434
9. **Hassan A. Mahmoud**, M. Shazly, Y. Bahie El-din, E. El-kashif, "Analysis of Composite Bolted Connection Joints under Out-of-Plane Loading," *Proc. ASME IMECE 2020*. DOI: 10.1115/IMECE2020-23303

Patents

All listed patents are provisional U.S. applications (submitted).

1. Y. Shi, **Hassan A. Mahmoud**, G. Lubineau, "Embedded Passive Wireless Capacitive Sensing System and Method for Real-Time Curing Monitoring", Docket No. 0338-855/2026-180-01, filed July 6, 2026.
2. A. Wagih, H. Ejaz, **Hassan A. Mahmoud**, G. Lubineau, "Inspection of GFRP Composite Adhesive Joint using Optical Transillumination", Docket No. 0338-843/2026-148-01, filed April 22, 2026.
3. **Hassan A. Mahmoud**, N. Jaber, G. Lubineau, "Wireless Capacitive Sensor for Low Frequency Vibrations," Docket No. 158982-613140/SA 61314 PA.
4. **Hassan A. Mahmoud**, O. Khalifa, M. Abdullah, A. H. Seno, N. Kouzahya, A. Asiri, A. Al-Jarro, A. Shehri, H. Saiari, T. Al-Naffouri, G. Lubineau, "Capacitive Strain Sensing IoT Device with Secure On-Demand Wireless Data Transmission," Docket No. 38136-3170001/SA 73651, filed November 15, 2025.
5. **Hassan A. Mahmoud**, O. Khalifa, M. Abdullah, A. H. Seno, N. Kouzahya, A. Asiri, A. Al-Jarro, A. Shehri, H. Saiari, T. Al-Naffouri, G. Lubineau, "Data Aggregator for Reading Sensed IoT Data on Demand", Docket No. 38136-3170001/SA 73658, filed November 15, 2025.
6. N. Divakaran, **Hassan A. Mahmoud**, G. Lubineau, "PDMS-MWCNT-Based Piezoresistive Ink with Tunable Viscosity for Stretchable Sensor Fabrication", No. 2026-029, filled September, 15, 2025.
7. **Hassan A. Mahmoud**, G. Lubineau, "Controlling the Sensitivity and Linearity of Capacitive Strain Sensors Using Non-Uniform Interdigitated Electrodes," Provisional U.S. Application No. 63/725,625, filed November 27, 2024.
8. A. Wagih, **Hassan A. Mahmoud**, R. Tao, G. Lubineau, "Toughening Thermoplastic Adhesive Tapes using Tailored Defects," Provisional U.S. Application No. 338-698/2022-0921-01, filed August 31, 2022.

Teaching & Mentorship

[ME 205] Basic Laboratory Skills Course

Mechanical Engineering, KAUST

Fall 2023, Spring 2024, Fall 2024

- Mentored master's students in experimental research methods and core laboratory practices.
- Instructed students on CAD and fabrication workflows, including 3D printing and laser cutting.
- Trained students in microscopy, image processing, oscilloscope operation, high-speed imaging, mechanical testing, strain gauge implementation, and material characterization.

MENTORSHIP

Co-mentoring two PhD students

KAUST

Ongoing

- Guided one PhD project on capacitive sensing for in-situ curing and aging monitoring of polymer composites.
- Guided a second PhD project on processing-structure-property relationships in laser-induced graphene.

Mentored two undergraduate research interns

KAUST

Completed

- Supervised research on laser-induced graphene-based flexible and stretchable strain/temperature sensors from design to validation.

Peer Review Activities

Reviewer for leading journals including:

- ACS Applied Electronic Materials (ACS)
- Carbon (Elsevier)
- Carbon Letters (Springer Nature)
- Surfaces and Interfaces (Elsevier)
- Advanced Engineering Materials (Wiley)

Conferences & Presentations

2026	European Workshop on SHM – EWSHM 2026 (Oral + Demo Session)	<i>Toulouse, France</i>
2025	MECC 19	<i>Dhahran, KSA</i>
2025	IEEE PIMRC (Oral + Demo Session)	<i>Istanbul, Turkey</i>
2025	SPIE Smart Structures + NDE 2025 (Oral and Poster)	<i>Vancouver, Canada</i>
2025	LOPEC 2025 (Poster)	<i>Munich, Germany</i>
2024	IEEE FLEPS 2024 (Poster)	<i>Tampere, Finland</i>
2024	European Workshop on SHM – EWSHM 2024 (Oral)	<i>Potsdam, Germany</i>
2024	LOPEC 2024 (Poster)	<i>Munich, Germany</i>
2023	Int. Workshop on SHM – IWSHM 2023 (Oral)	<i>Stanford, USA</i>
2023	ECCOMAS SMART 2023 (Oral)	<i>Patras, Greece</i>
2023	Energizing the Future with Composites (Poster)	<i>KAUST, KSA</i>
2022	European Workshop on SHM – EWSHM 2022 (Oral)	<i>Palermo, Italy</i>
2020	ASME IMECE 2020 (Oral, Virtual)	<i>Virtual</i>

Courses & Workshops

2026	The Era of AI and Digitalization for Structural Applications, TU Delft	<i>The Netherlands</i>
2023	SHM for Integrity Management of Civil Structures and Infrastructures, Politecnico di Milano	<i>Italy</i>
2022	Basic Electromagnetic FEA, NAFEMS	<i>Virtual</i>
2022	SHM using Statistical Pattern Recognition, Los Alamos Dynamics, LLC	<i>Virtual</i>
2022	Composites Design Workshops XXII, Stanford University	<i>Virtual</i>

References

Gilles Lubineau

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Ahmed Wagih

RESEARCH SCIENTIST, PHYSICAL SCIENCE AND ENGINEERING DIVISION, KAUST

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Tareq Al-Naffouri

PROFESSOR, COMPUTER, ELECTRICAL AND MATHEMATICAL SCIENCE AND ENGINEERING DIVISION, KAUST

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Nizar Jaber

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